

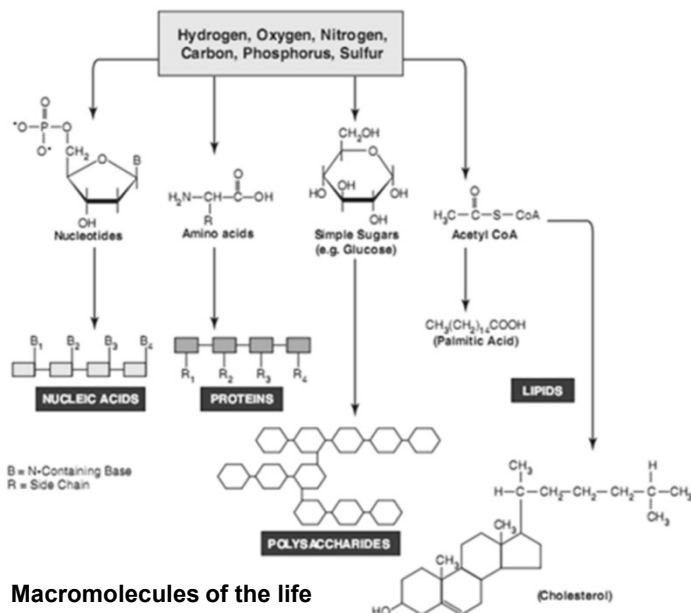
Biomolecules

Proteins



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Biomolecules



Proteins

- ❖ Term Protein is derived from PROTEOSE- “**of the 1st Rank**”. They are polypeptide chains consists of several AA which are linked together with peptide bond, a process called **Polymerization**.
- ❖ Proteins are organic molecules consisting of multiple amino acids thus are biopolymers of amino acids.
- ❖ Proteins differ from one another primarily in their sequence of amino acids, which is governed by the nucleotide sequence of their genes, and which usually results in protein folding into a specific 3-dimensional structure that determines its activity.
- ❖ The polypeptide will always have **NH₂** at one end, **the N-terminus** and **COOH** at the other, the **C-terminus**.
- ❖ The nature of the 'R' group determines how **hydrophilic** or **hydrophobic** the amino acid is.
- ❖ There are **total 20-amino acids** which involve in the formation of proteins in living systems.

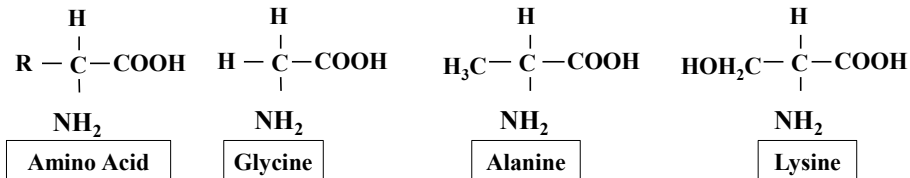
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There are 22 genetically encoded (proteinogenic) amino acids, 20 in the standard genetic code and an additional 2 (selenocysteine and pyrrolysine) that can be incorporated by special translation mechanisms.

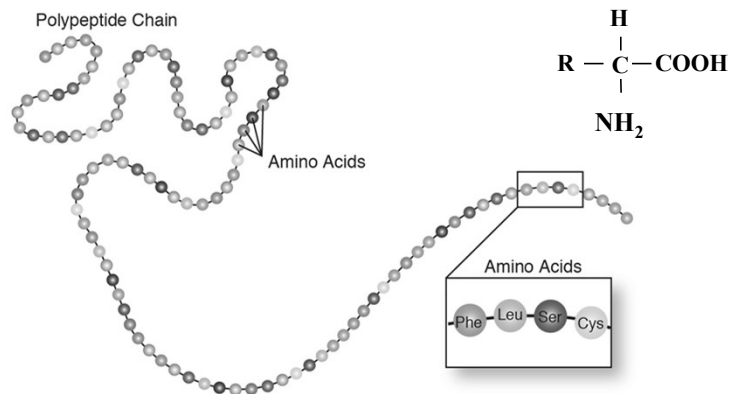
Amino Acids

- ❖ There are 20 different amino acids found in nature. Amino acids are building block of all the proteins molecules.
- ❖ AAs are the monomeric building blocks of proteins. They are composed of both acids and amines and they have a central Carbon atom to which a carboxyl group, an amino group, a H-atom and one variable-R side up or side chain is attached.



- ❖ **R-group:** It is variable and we have 20 known R-groups therefore, we have only 20 AA.
- ❖ The sequence of joined amino acids is called as **Peptide** - Dipeptide (2), Tri-peptide (3), Oligopeptide (3-10), Polypeptide (more than 10).
- ❖ **N- and C-terminus.**

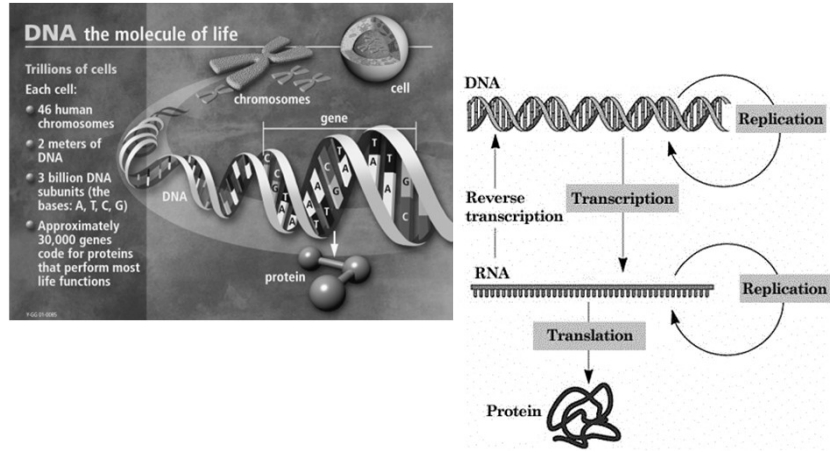
Amino Acid Chain



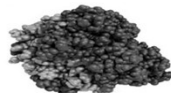
Amino Acids

Ala: Alanine	Gln: Glutamine	Leu: Leucine	Ser: Serine
Arg: Arginine	Glu: Glutamic acid	Lys: Lysine	Thr: Threonine
Asn: Asparagine	Gly: Glycine	Met: Methionine	Trp: Tryptophane
Asp: Aspartic acid	His: Histidine	Phe: Phenylalanine	Tyr: Tyrosine
Cys: Cysteine	Ile: Isoleucine	Pro: Proline	Val: Valine

Central Dogma of Molecular Biology

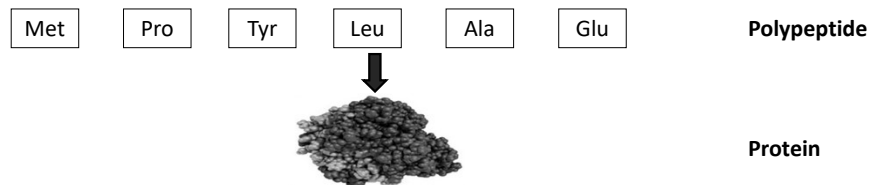


Enzymes

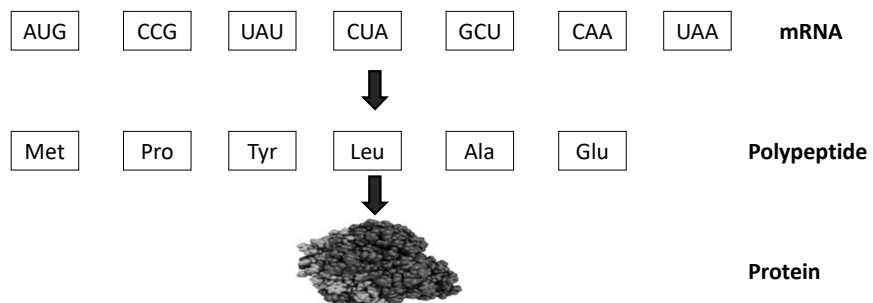


Protein

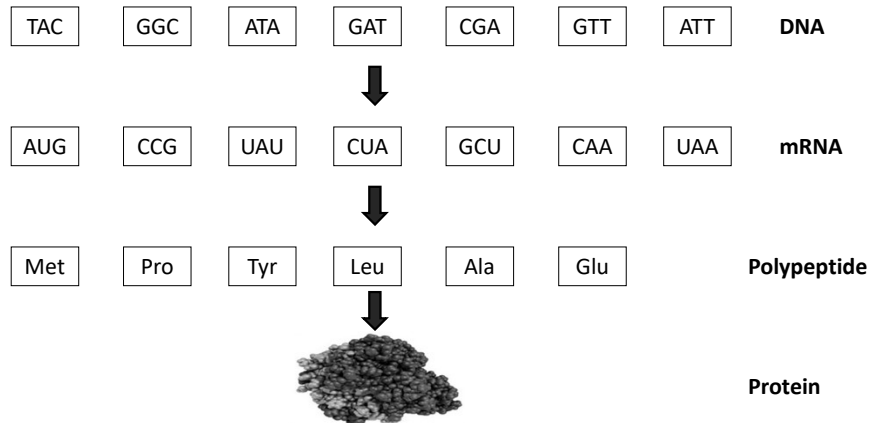
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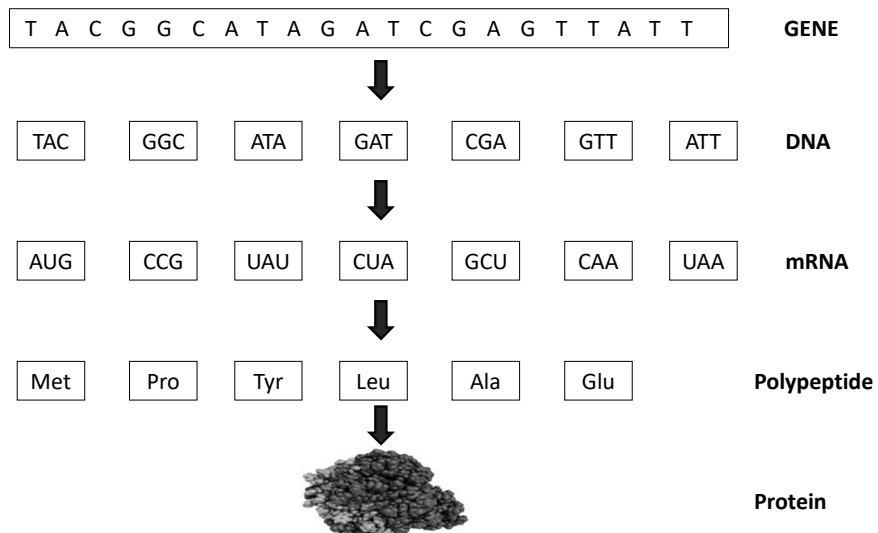
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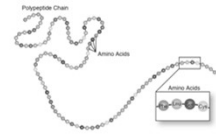
Enzymes



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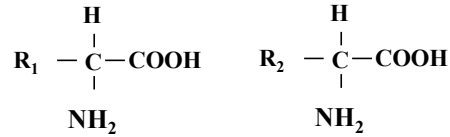


Peptide Bond / Linkage

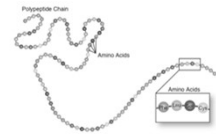


Amino Acids

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Asp Aspartic acid	His Histidine	Phe Phenylalanine	Tyr Tyrosine
Cys Cysteine	Ile Isoleucine	Pro Proline	Val Valine

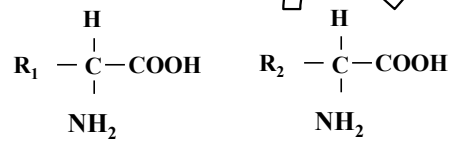


Peptide Bond / Linkage



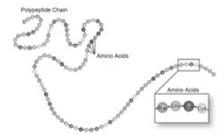
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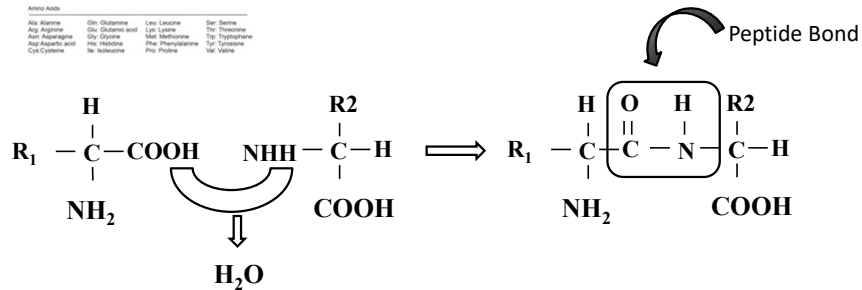


Rotating 90° clockwise
 $\text{NH}_2 = \text{NHH}$

Peptide Bond / Linkage



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Cys: Cysteine	Ile: Isoleucine	Pro: Proline	Val: Valine



- ✓ Two amino acids are linked together through carboxylic and amino group to produce the primary structure of a protein chain. The bond between the two adjacent AA is called **Peptide Bond** that forms a peptide chain.

Classification of Amino Acids

1. Non-polar or Hydrophobic R-group:

They contain both aliphatic and aromatic side chains which are hydrophobic in nature e.g. alanine, valine, leucine, isoleucine, proline, Phenyl alanine, tryptophan, methionine.

2. Polar but charged R-groups:

They contain polar side chain residues that can participate in H-bond formation e.g. serine, threonine, tyrosine, cystine, glutamine, asparagine.

3. Positively charged R-groups:

In this, the R-groups are positively charged e.g. Histidine, lysine, arginine.

4. Negatively charged R-groups:

In this, the R-groups are negatively charged e.g. Aspartic acid, glutamic acid.

- ✓ **Essential amino acids:** They cannot be made by the body. As a result, they must come through diet (outside) e.g. histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.
- ✓ **Non-essential amino acids:** They are required for normal health and growth, and can be synthesized within the body or derived in the body from essential amino acids e.g. alanine, asparagine, aspartic acid, cystine, glutamic acid, glutamine, glycine, proline, serine, and tyrosine

Protein structure

1. Primary structure:

In this the linear sequence of AA in a polypeptide chain linked together with peptide bonds and no other bonds or forces are included e.g. Insulin.

2. Secondary structure:

It refers to a regular folding patterns of continuous portions of polypeptide chain. This arrangement is maintained by **non-covalent links** called H-bonds. 2-types of folding and coiling occurs in this -

a) α -Helix: When the polypeptide is arranged in a spiral or spring like manner and the loops are held together by H-bonds e.g. Keratine.

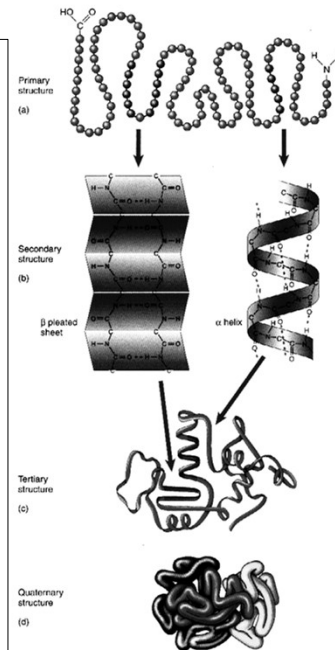
b) β -Pleated sheet: Two or more polypeptide chains are attached together by intermolecular H-bonds and the chains are parallel or antiparallel e.g. Silk.

3. Tertiary structure:

Further folding or coiling of secondary structure will give rise to tertiary structure and it forms a 3-D stable structure. This will involve formation of ionic bonds, H-bonds and disulphide bonds e.g. Lysozyme.

4. Quaternary structure:

It is the association of many polypeptide units. Each unit develops its tertiary structure and functions as a sub-unit of protein. Different sub-units pack together to give quaternary conformation e.g. Hemoglobin.



Functions or Biological Significance of Proteins

❖ After carbohydrates, proteins are essential biomolecules for living beings. They often constitute the majority share (60%) of the dry weight of cells. They are vital chemical substances which are essential for the growth and maintenance of the life. The main sources of protein are milk, cheese, pulses, fish etc.

- 1. Enzyme catalysis** - facilitates/speeds up bio-chemical reactions e.g. enzymes. Amylase converts starch to simple sugar.
- 2. Defense** - recognizes foreign microbes; forms the center of the immune system e.g. immunoglobulins, toxins, antibodies (Globular proteins that "recognize" foreign microbes). Fibrous proteins like keratin forms nails, horns, hoof, hairs, scales, feathers which help to provide protection.
- 3. Transport** - moves certain small molecules/ions e.g. hemoglobin, proton pump. Hemoglobin is red blood cell protein, Genetic information transfer - nucleoproteins.
- 4. Structure/support** - structural role e.g. fibers, collagen (most abundant protein in vertebrates), keratin, fibrin. Collagen forms the matrix of skin, ligaments, tendons and bones, and membranes.
- 5. Motion** - a muscle protein responsible for muscle contraction contracting muscles e.g. actin, myosin.
- 6. Regulation** - receives/sends information to regulate body functions e.g. hormones. Hormones serve as intercellular messengers e.g. Insulin (blood sugar regulation).
- 7. Storage** - Occurs in Milk-Casein, Egg-Ovalbumin, Seeds-Gluten.

THANK YOU